Malware Analysis Report: Overlay Payload Extraction

Author: Ahmed Emad Nasr **Date:** April 7, 2026

Category: Malware Analysis / Reverse Engineering

Executive Summary

This report details the static analysis of a suspicious executable file disguised as a PDF document. The investigation revealed that the initial file acts as a loader, utilizing multiple obfuscation techniques to evade detection. The primary malicious payload (Stage 2) was discovered hidden within the file's Overlay section, padded with null bytes to break standard signature detection. Manual extraction and reconstruction of the PE file exposed a secondary executable utilizing timestomping and high entropy, confirming its malicious nature.

Indicators of Compromise (IoCs)

Based on the provided static analysis, the following hashes were identified:

- **Original Dropper/Loader (Stage 1):**
  - **Filename:** singlefilehost.exe
  - **SHA-256:** DCE2488DDC3AB45FE8DF1C2BB3E21CD11D4E697E94E0E2A88B4BFD35C6516055
- **Extracted Payload (Stage 2):**
  - **Filename:** overlay.dump / tkfgk435jkdgf.dll
  - **SHA-256:** 160D0260FD2DF7DA4381AB01E87D7C047D4A381C59D35FE1B60F0ABE162A915A

Phase 1: Initial Static Analysis (Stage 1)

Initial inspection of the target file (singlefilehost.exe) was conducted using **PeStudio**, revealing several critical anomalies and indicators of malicious intent:

1. Social Engineering & Masquerading

- **Icon Masking:** The executable utilizes a standard PDF file icon to deceive the user into believing it is a legitimate document.

- **Digital Certificate Anomalies:** The file is signed with a legitimate but expired Microsoft certificate (Microsoft Code Signing PCA 2010, valid from Jan 2022 to Jun 2023). This technique is often used to bypass rudimentary security checks that blindly trust digitally signed binaries.

2. File Structure Anomalies

- **Suspicious Original Filename:** The version information reveals an original filename of tkfgk435jkdgf.dll, which consists of random characters, a common trait of packer or builder tools.
- **Abnormal Section Permissions:** The PE headers show multiple sections with Execute permissions (e.g., .text and .CLR_UEF). A standard, uncompiled executable typically contains only one executable section containing the code. This indicates a packed or self-extracting architecture.

3. Payload Concealment (Overlay & Resources)

- **Massive Overlay:** The most significant finding was an Overlay section comprising **87.52%** of the total file size.
- **Executable Resource:** A resource named MINIDUMP_... was identified as a 64-bit executable file.
- The overall high entropy combined with the massive overlay strongly suggested that the actual malicious payload was appended to the end of the file rather than being contained within the standard PE sections.

Phase 2: Advanced Static Analysis & Manual Unpacking

To investigate the contents of the hidden data, the Overlay section was dumped to a separate file (overlay.dump) for manual analysis using **HxD (Hex Editor)**.

Obfuscation via Null Byte Padding

- Initial inspection of the dumped overlay did not immediately reveal standard PE magic bytes at offset 0x00.
- Upon closer inspection within the hex editor, the standard DOS header magic bytes 4D 5A (MZ) were located further down in the hex dump.
- **Evasion Technique:** The malware author prepended numerous Null Bytes (00) to the beginning of the embedded executable. This is a deliberate technique designed

to break automated static analysis tools and simplistic YARA rules that expect the MZ header to be at the exact beginning of the file.

Payload Reconstruction

- The prepended null bytes were manually deleted using the hex editor, shifting the MZ header to the beginning of the file (offset 0x00), thereby successfully reconstructing the embedded PE file for further analysis.

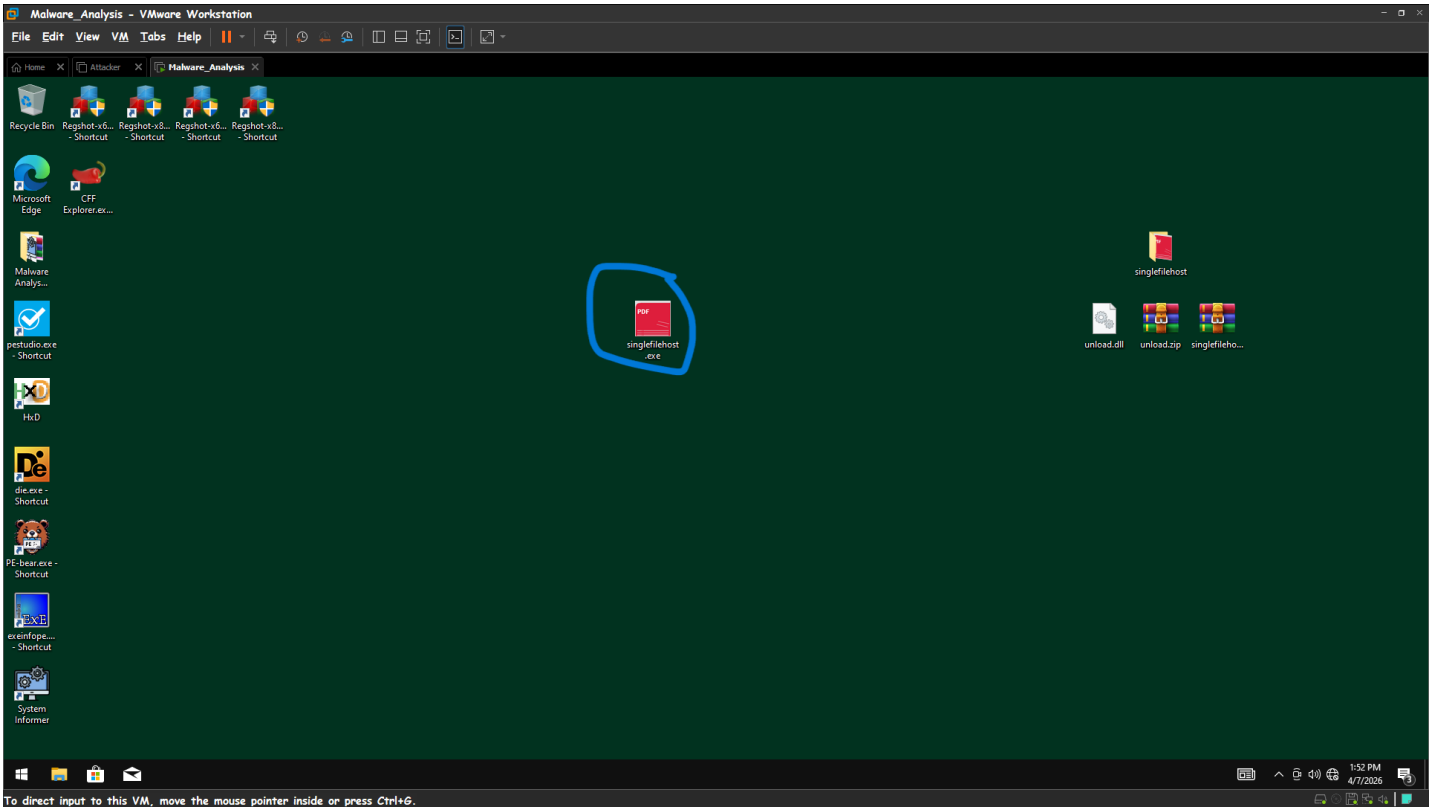
Phase 3: Stage 2 Payload Analysis

The reconstructed executable (Stage 2 payload) was re-analyzed in PeStudio, revealing further malicious artifacts:

- **Timestomping:** The compilation timestamp in the PE header was explicitly altered to a future date (Thu Nov 10 05:28:50 2061 UTC). This is a known anti-forensics technique used to disrupt timeline analysis and bypass age-based heuristics.
- **Debug Artifacts:** A hardcoded debug path was extracted: C:\Users\admin\source\repos\SDCBundle\SDCBundle\obj\Release\net6.0-win-x64\tkfgk435jkdgf.pdb This provides valuable intelligence regarding the malware author's development environment (project name SDCBundle, user admin, compiled for .NET 6.0 x64).
- **High Entropy:** The extracted payload maintains a high entropy (6.730), suggesting that this stage might still contain encrypted configurations or further packed components.

Conclusion

The analyzed file is a sophisticated **Loader/Dropper**. It relies on social engineering (PDF masquerading) and certificate abuse to achieve initial execution. Its primary function is to unpack and execute a hidden Stage 2 payload located in its overlay section. The malware demonstrates clear evasion techniques, specifically null-byte padding to obscure the embedded PE header and timestomping to frustrate forensic analysis.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Recycle Bin
Regshot-x6... - Shortcut
Regshot-x8... - Shortcut
Regshot-x6... - Shortcut
Regshot-x8... - Shortcut
Microsoft Edge
CFF Explorer.ex...
Malware Analys...
pestudio.exe - Shortcut
HxD
die.exe - Shortcut
PE-bear.exe - Shortcut
exeinfope.... - Shortcut
System Informer
singlefilehost.exe
singlefilehost
unload.dll
unload.zip
singlefileho...
1:52 PM
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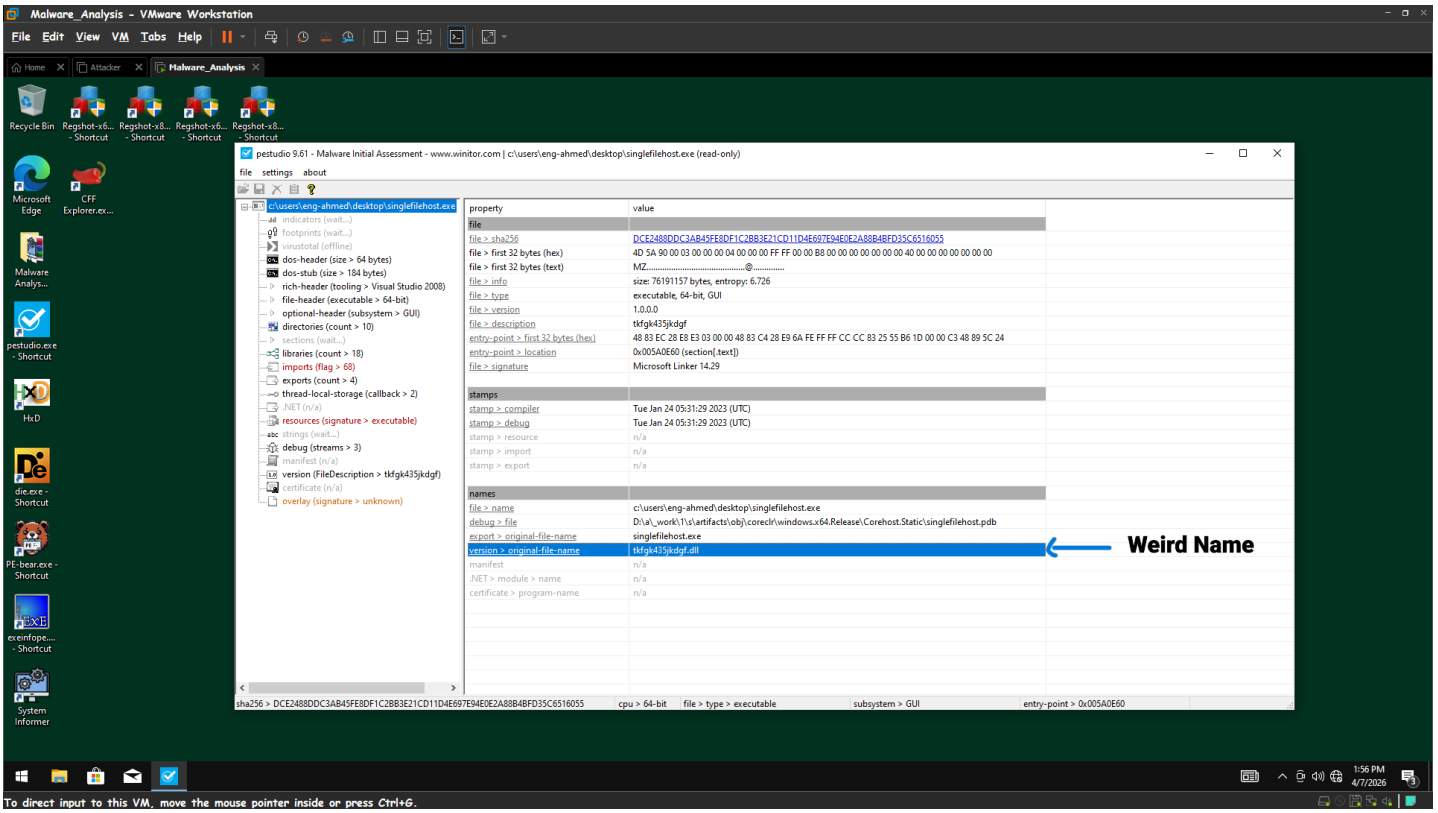
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.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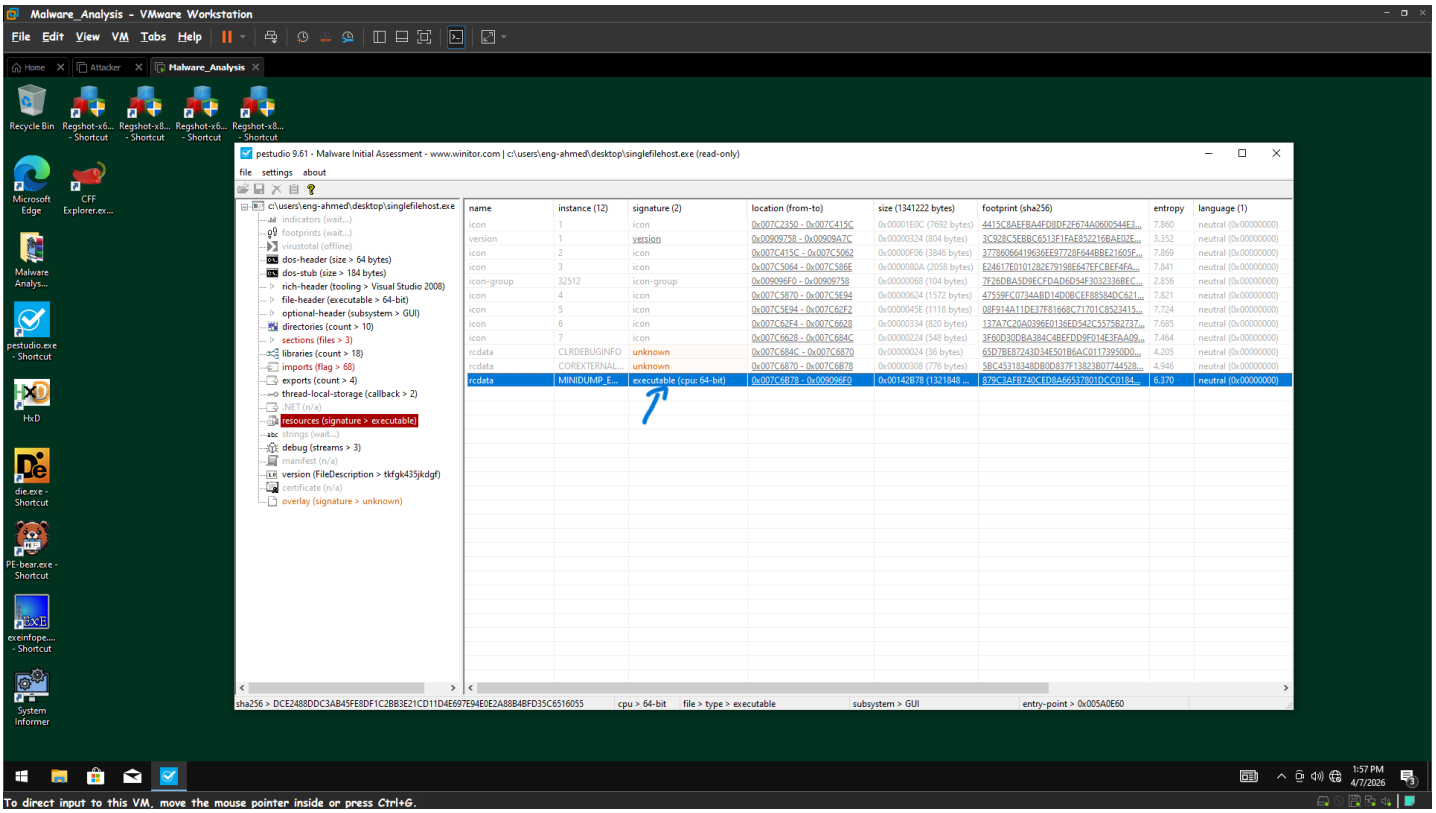
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\singlefilehost.exe (read-only)
file settings about
c:\users\eng-ahmed\desktop\singlefilehost.exe
indicators (wait...)
footprints (wait...)
virustotal (offline)
dos-header (size > 64 bytes)
dos-stub (size > 184 bytes)
rich-header (tooling > Visual Studio 2008)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 10)
sections (wait...)
libraries (count > 18)
imports (flag > 68)
exports (count > 4)
thread-local-storage (callback > 2)
.NET (n/a)
resources (signature > executable)
strings (wait...)
debug (streams > 3)
manifest (n/a)
version (FileDescription > tkfgk435jkdgf)
certificate (n/a)
overlay (signature > unknown)
property value
file
file > sha256 DCE2488DDC3AB45FE8DF1C2BB3E21CD11D4E697E94E0E2A88B4BFD35C6516055
file > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text) MZ.....@.....
file > info size: 76191157 bytes, entropy: 6.726
file > type executable, 64-bit, GUI
file > version 1.0.0.0
file > description tkfgk435jkdgf
entry-point > first 32 bytes (hex) 48 83 EC 28 E8 E3 03 00 00 48 83 C4 28 E9 6A FE FF FF CC CC 83 25 55 B6 1D 00 00 C3 48 89 5C 24
entry-point > location 0x005A0E60 (section[.text])
file > signature Microsoft Linker 14.29
stamps
stamp > compiler Tue Jan 24 05:31:29 2023 (UTC)
stamp > debug Tue Jan 24 05:31:29 2023 (UTC)
stamp > resource n/a
stamp > import n/a
stamp > export n/a
names
file > name c:\users\eng-ahmed\desktop\singlefilehost.exe
debug > file D:\a_work\1\s\artifacts\obj\coreclr\windows.x64.Release\Corehost.Static\singlefilehost.pdb
export > original-file-name singlefilehost.exe
version > original-file-name tkfgk435jkdgf.dll
manifest n/a
.NET > module > name n/a
certificate > program-name n/a
Weird Name
sha256 > DCE2488DDC3AB45FE8DF1C2BB3E21CD11D4E697E94E0E2A88B4BFD35C6516055
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x005A0E60
1:56 PM
4/7/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.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file settings about
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libraries (count > 18)
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thread-local-storage (callback > 2)
.NET (n/a)
resources (signature > executable)
strings (wait...)
debug (streams > 3)
manifest (n/a)
version (FileDescription > tkfgk435jkdgf)
certificate (n/a)
overlay (signature > unknown)
name | instance (12) | signature (2) | location (from-to) | size (1341222 bytes) | footprint (sha256) | entropy | language (1)
icon | 1 | icon | 0x007C2350 - 0x007C415C | 0x00001E0C (7692 bytes) | 4415C8AEFBA4FD8DF2F674A0600544E3... | 7.860 | neutral (0x00000000)
version | 1 | version | 0x00909758 - 0x00909A7C | 0x00000324 (804 bytes) | 3C928C5EBBC6513F1FAE852216BAE02E... | 3.352 | neutral (0x00000000)
icon | 2 | icon | 0x007C415C - 0x007C5062 | 0x00000F06 (3846 bytes) | 377860664193636EE97728F644BBE21605F... | 7.869 | neutral (0x00000000)
icon | 3 | icon | 0x007C5064 - 0x007C586E | 0x0000080A (2058 bytes) | E24617E010128 2E79198E647EFCBEF4FA... | 7.841 | neutral (0x00000000)
icon-group | 32512 | icon-group | 0x009096F0 - 0x00909758 | 0x00000068 (104 bytes) | 7F26DBA5D9ECFDAD6D54F3032336BEC... | 2.856 | neutral (0x00000000)
icon | 4 | icon | 0x007C5870 - 0x007C5E94 | 0x00000624 (1572 bytes) | 47559FC0734ABD14D0BCEF88584DC621... | 7.821 | neutral (0x00000000)
icon | 5 | icon | 0x007C5E94 - 0x007C62F2 | 0x0000045E (1118 bytes) | 08F914A11DE37F81668C71701C8523415... | 7.724 | neutral (0x00000000)
icon | 6 | icon | 0x007C62F4 - 0x007C6628 | 0x00000334 (820 bytes) | 137A7C20A0396E0136ED542C5573B2737... | 7.685 | neutral (0x00000000)
icon | 7 | icon | 0x007C6628 - 0x007C684C | 0x00000224 (548 bytes) | 3F60D30DBA384C4BEFDD9F014E3FAA09... | 7.464 | neutral (0x00000000)
rcdata | CLRDEBUGINFO | unknown | 0x007C684C - 0x007C6870 | 0x00000024 (36 bytes) | 65D7BE87243D34E501B6AC01173950D0... | 4.205 | neutral (0x00000000)
rcdata | COREXTERNAL... | unknown | 0x007C6870 - 0x007C6B78 | 0x00000308 (776 bytes) | 5BC45318348DB0D837F13823B077445 28... | 4.946 | neutral (0x00000000)
rcdata | MINIDUMP_E... | executable (cpu: 64-bit) | 0x007C6B78 - 0x009096F0 | 0x00142B78 (1321848 ... | 879C3AFB740CED8A6653780 1DCC0184... | 6.370 | neutral (0x00000000)
sha256 > DCE2488DDC3AB45FE8DF1C2BB3E21CD11D4E697E94E0E2A88B4BFD35C6516055
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x005A0E60
1:57 PM
4/7/2026
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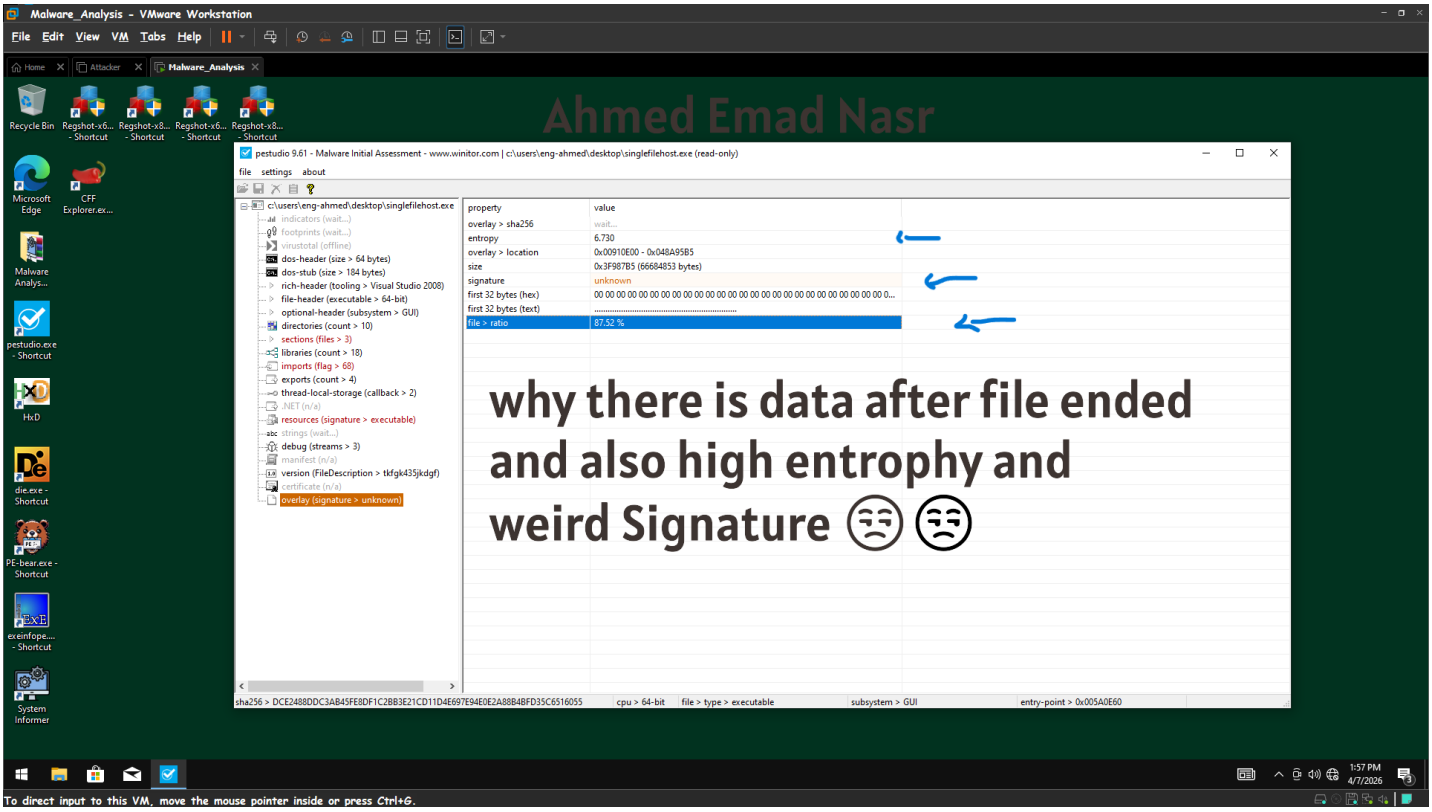
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manifest (n/a)
version (FileDescription > tkfgk435jkdgf)
certificate (n/a)
overlay (signature > unknown)
property
value
overlay > sha256
wait...
entropy
6.730
overlay > location
0x00910E00 - 0x048A95B5
size
0x3F987B5 (66684853 bytes)
signature
unknown
first 32 bytes (hex)
first 32 bytes (text)
file > ratio
87.52 %
why there is data after file ended and also high entrophy and weird Signature 😒😒
sha256 > DCE2488DDC3AB45FE8DF1C2BB3E21CD11D4E697E94E0E2A88B4BFD35C6516055
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x005A0E60
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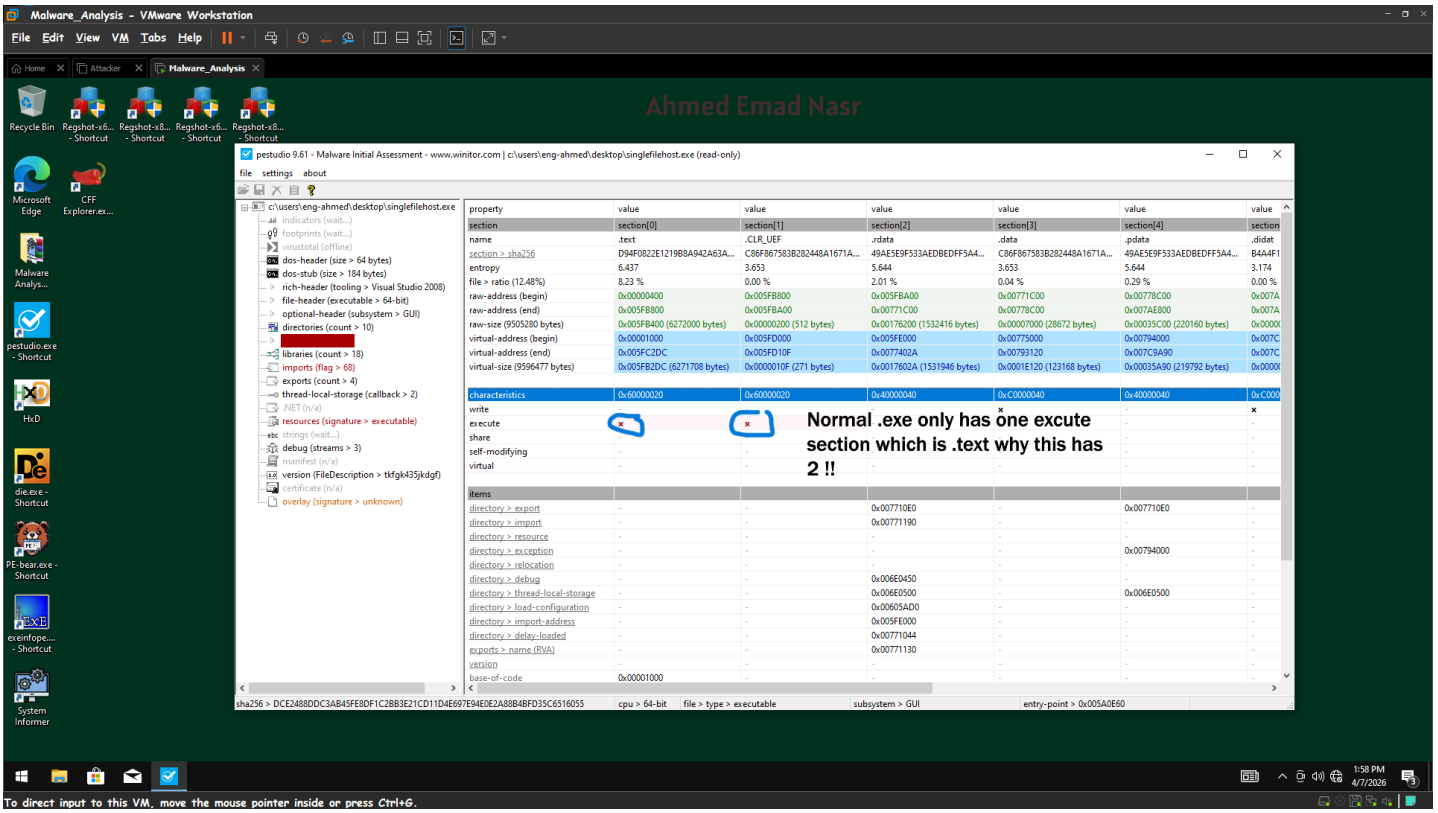
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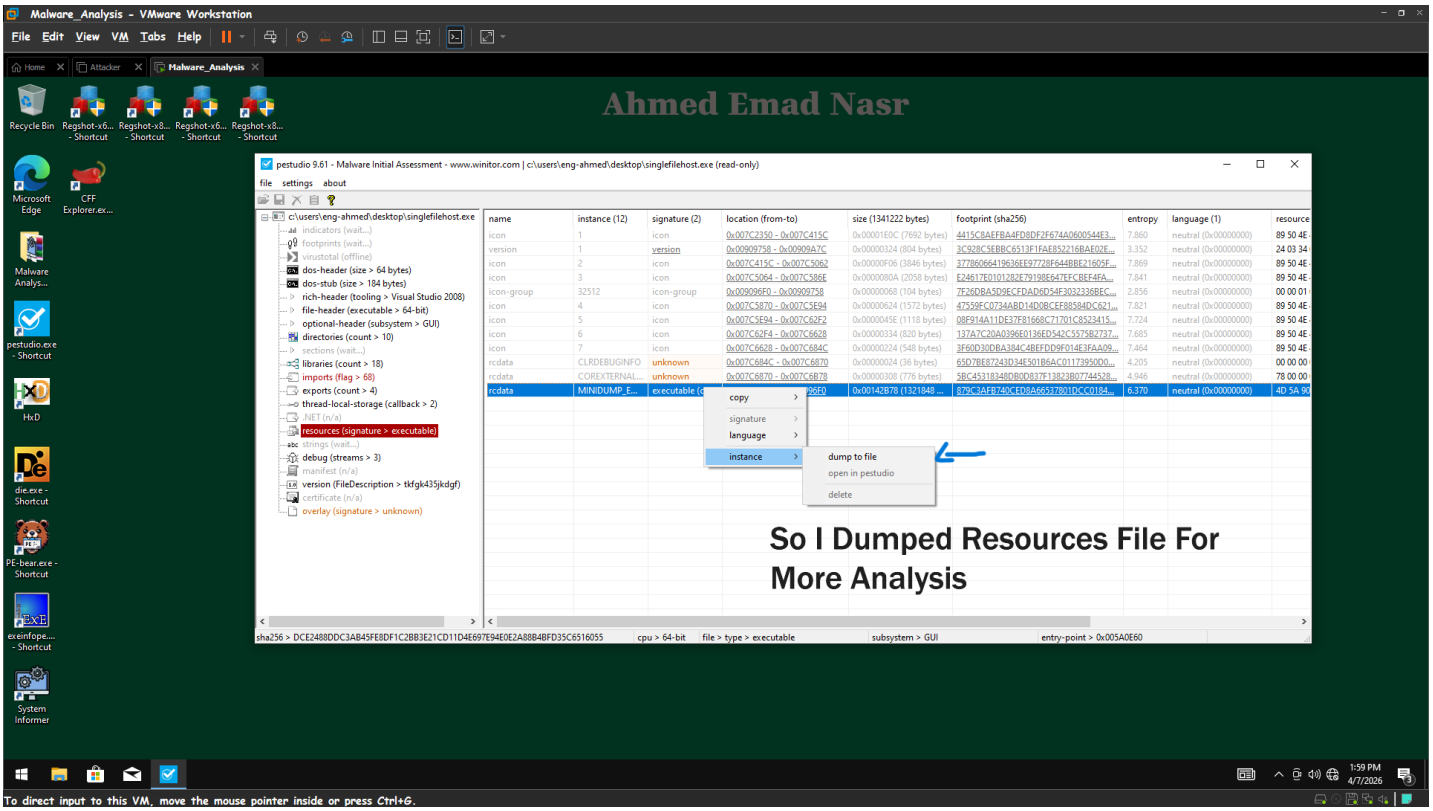
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\singlefilehost.exe (read-only)
Normal .exe only has one excute section which is .text why this has 2 !!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.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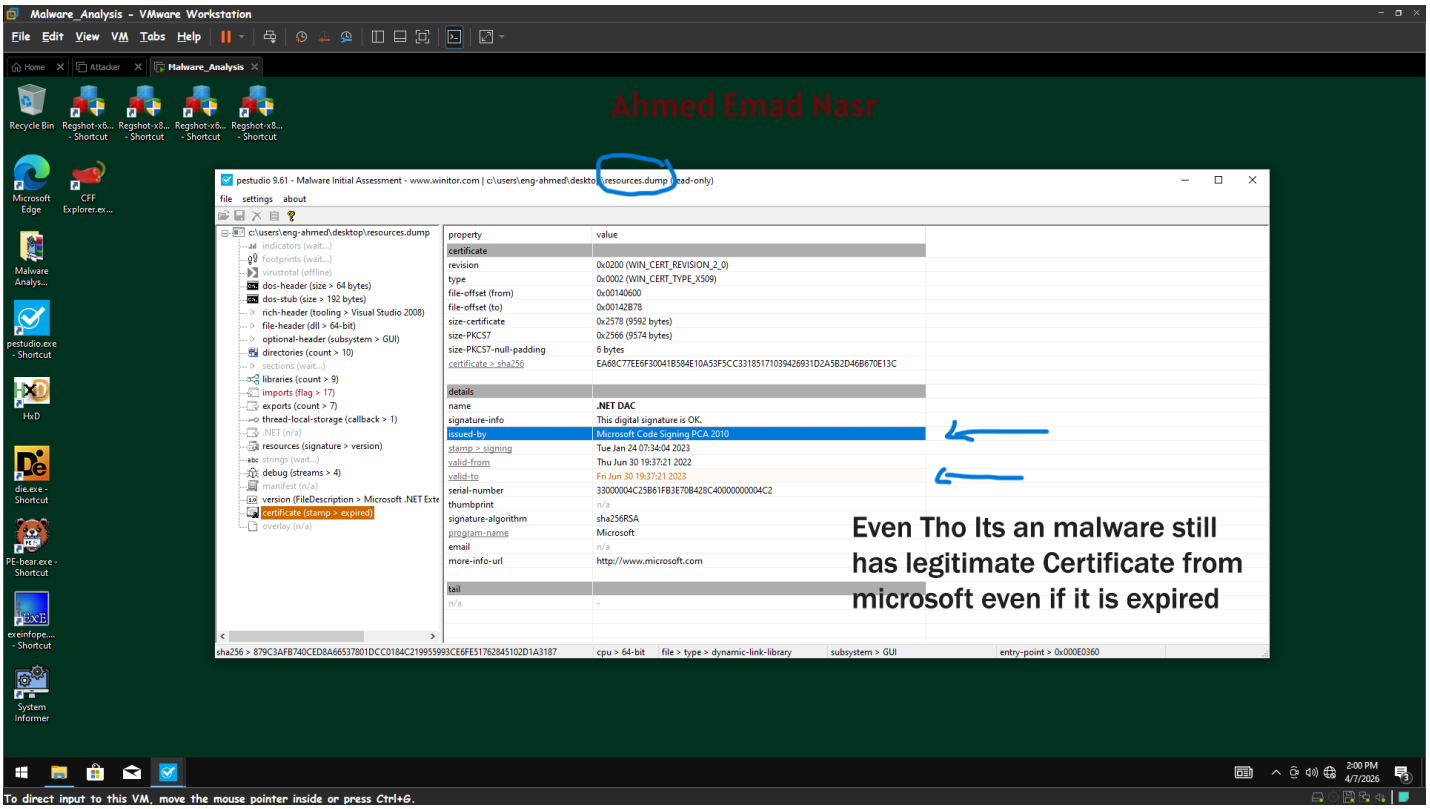
So I Dumped Resources File For More Analysis
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
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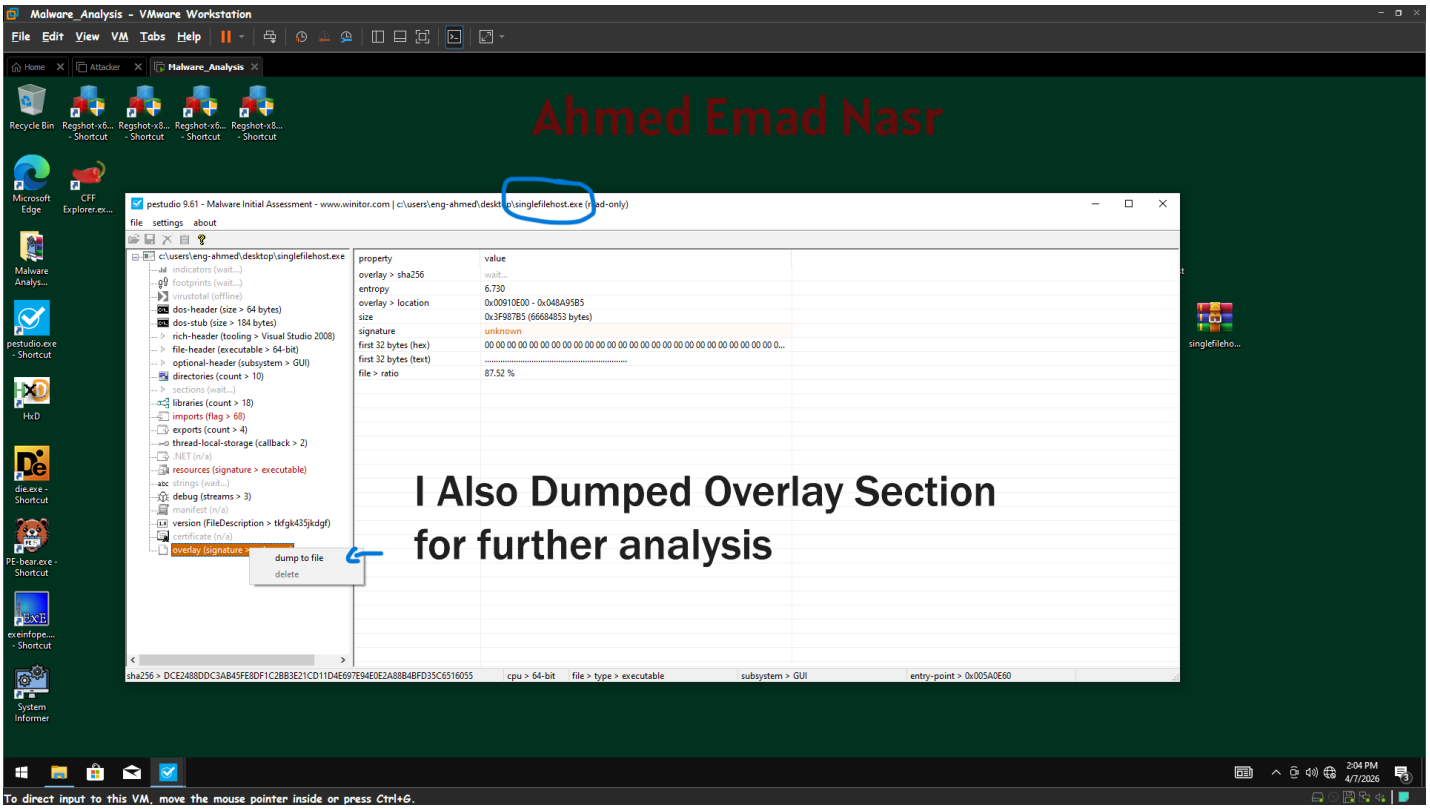
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\resources.dump (read-only)
Even Tho Its an malware still has legitimate Certificate from microsoft even if it is expired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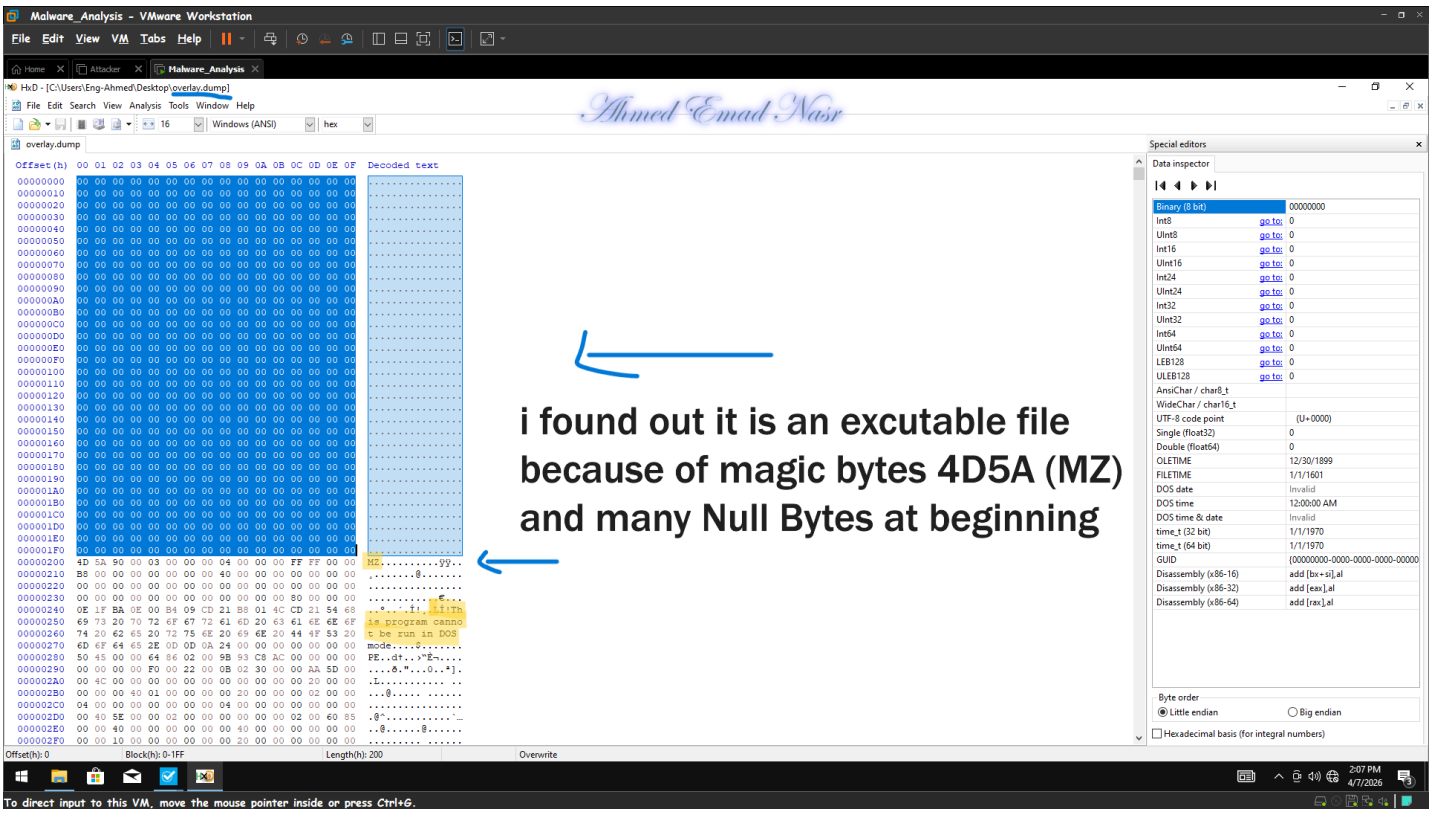
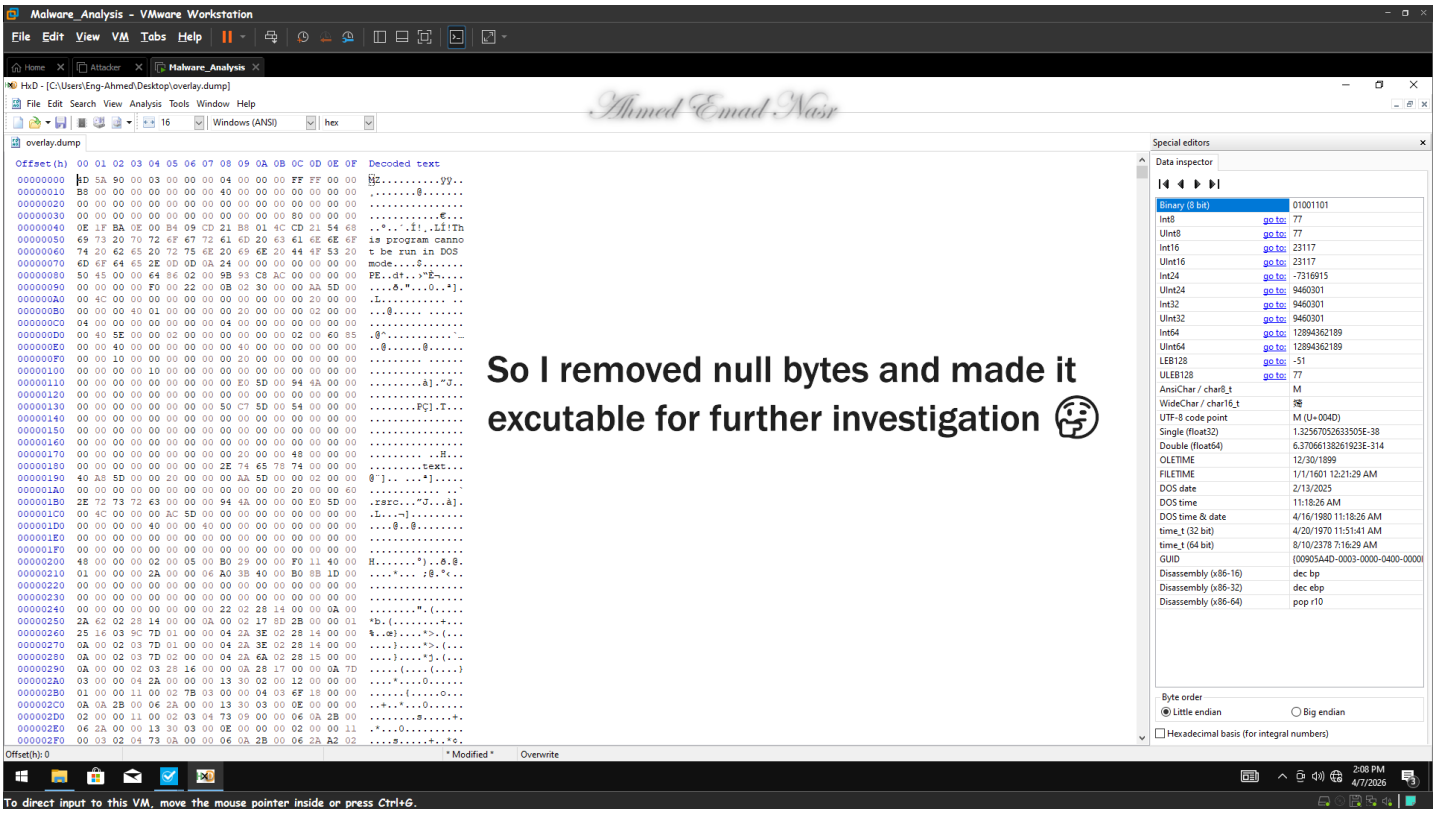
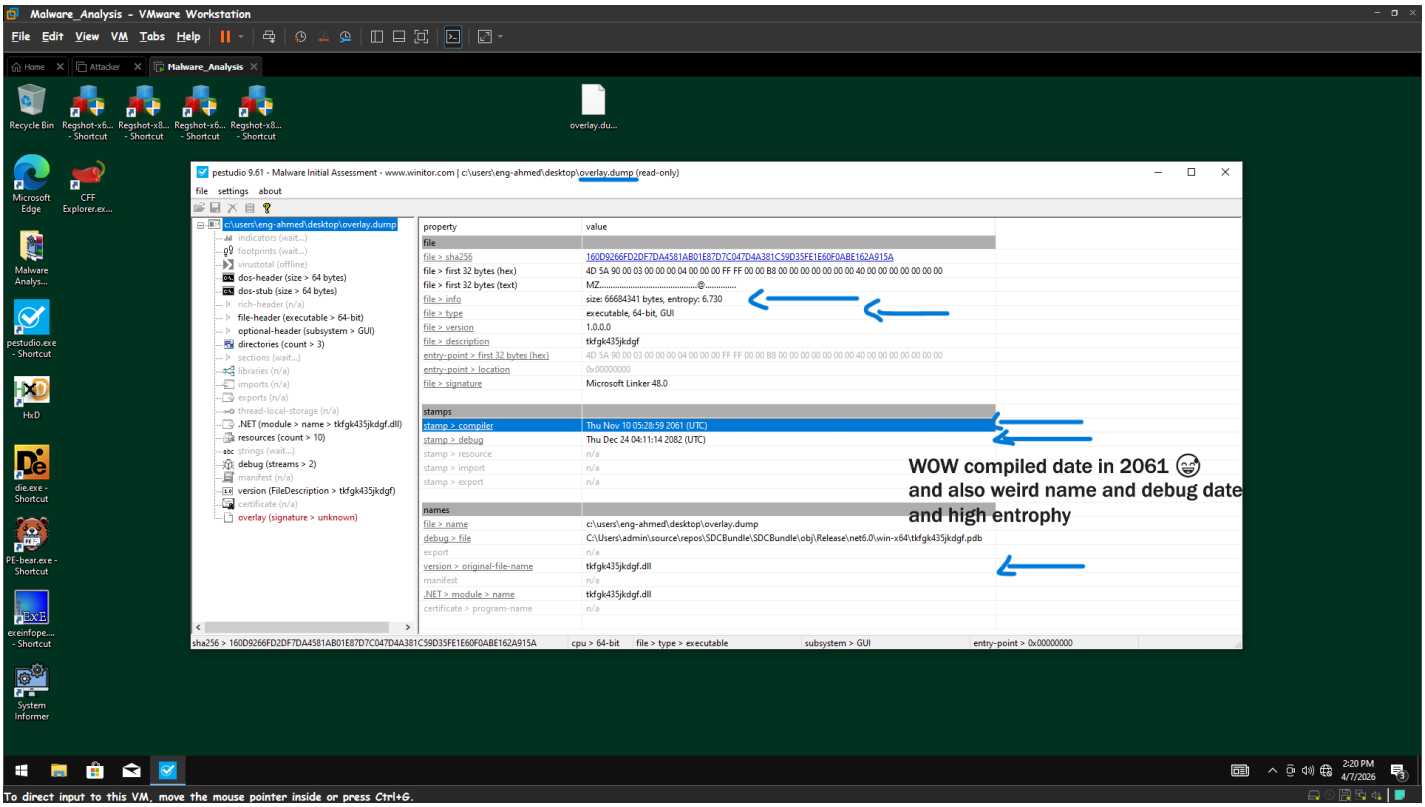
I Also Dumped Overlay Section for further analysis

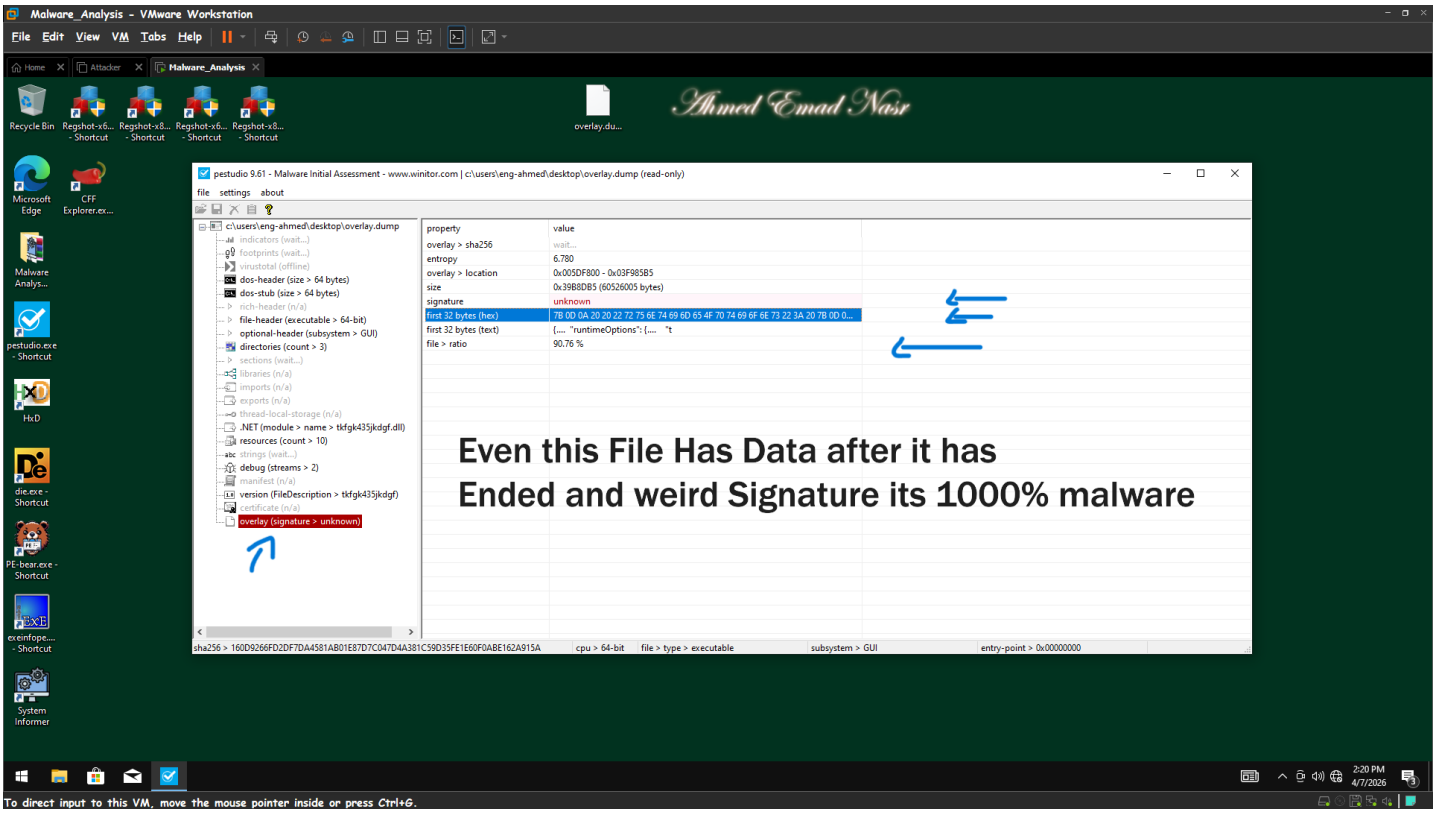


i found out it is an excutable file because of magic bytes 4D5A (MZ) and many Null Bytes at beginning



So I removed null bytes and made it excutable for further investigation 🤔


Malware_Analysis - VMware Workstation
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\overlay.dump (read-only)
file > sha256 160D9266FD2DF7DA4581AB01E87D7C047D4A381C59D35FE1E60F0ABE162A915A
file > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text) MZ.....@.....
file > info size: 66684341 bytes, entropy: 6.730
file > type executable, 64-bit, GUI
file > version 1.0.0.0
file > description tkfgk435jkdgf
entry-point > location 0x00000000
file > signature Microsoft Linker 48.0
stamp > compiler Thu Nov 10 05:28:59 2061 (UTC)
stamp > debug Thu Dec 24 04:11:14 2082 (UTC)
file > name c:\users\eng-ahmed\desktop\overlay.dump
debug > file C:\Users\admin\source\repos\SDCBundle\SDCBundle\obj\Release\net6.0\win-x64\tkfgk435jkdgf.pdb
version > original-file-name tkfgk435jkdgf.dll
.NET > module > name tkfgk435jkdgf.dll
WOW compiled date in 2061 😜
and also weird name and debug date
and high entropy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overlay > sha256 wait...
entropy 6.780
overlay > location 0x005DF800 - 0x03F985B5
size 0x39B8DB5 (60526005 bytes)
signature unknown
first 32 bytes (hex) 7B 0D 0A 20 20 22 72 75 6E 74 69 6D 65 4F 70 74 69 6F 6E 73 22 3A 20 7B 0D 0...
first 32 bytes (text) {.... "runtimeOptions": {.... "t
file > ratio 90.76 %
overlay (signature > unknown)
Even this File Has Data after it has
Ended and weird Signature its 1000% malware